

AILT - Study 4 Post Test 2 - Capability (#150323)

Author(s)

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Pre-registered on: 2023/11/08 - 01:29 PM (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

We are examining the relationship between AI literacy and perceptions of AI's capability to complete tasks. Specifically, we show in another study that people with greater AI literacy have lower AI receptivity across a broad range of tasks. This study is a post-test to examine whether these results are the result of people with greater AI literacy simply perceiving AI as less capable of completing those tasks.

3) Describe the key dependent variable(s) specifying how they will be measured.

The dependent measure will ask participants "To what extent do you think that an AI algorithm is capable of completing each of the following tasks?" for 26 different tasks on a 7-point scale (1 = definitely incapable, 7 = definitely capable). The 26 tasks are those used in Castelo, Bos, & Lehmann (2019).

4) How many and which conditions will participants be assigned to?

There will be no conditions. Our independent variable will be participants' AI literacy as measured using a 25-item multiple choice test. The sum of the correct responses will serve as their knowledge level.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will calculate participants' average perception of AI's capability to complete tasks across the 26-tasks. We will then regress this average preference on their AI literacy score.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We include an attention check that asks people to answer the simple equation of 12-4. People who do not answer this correctly will be excluded from taking the survey.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We will post the survey for 200 participants.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will also examine the correlations between knowledge and capability perceptions across the 26 items individually.

We will collect gender and age.